

Introduction :

- Several variables were related with higher risk of mortality during the initial wave of the COVID-19 outbreak in Bangladesh, including age>65, density, sanitizing criteria, hospital facilities, urban population and housing conditions. In order to target control measures and plan for future waves of the epidemic, public health agencies need to understand how these vulnerabilities are distributed across and clustered within communities.

Objective:

- To examine the socioeconomic and demographic risk factors of the COVID-19 mortality in Bangladesh.

Methods:

- The number of COVID-19 cases are visualized in different district. Here, district wise secondary data are excerpted from reliable sources namely DGHS dashboard, BBS, HIES.
- For all the 64 districts of Bangladesh, we defined the outcome variable, COVID-19 mortality as the number of deaths in a given district per 100,000 population.
- Firstly, we pursued correlations and distributions of eight vulnerability measures.
- Then, we conducted a negative binomial with log link analysis to expected number of deaths.
- Because of the nature of dependent variable and overdispersion in our data is remarkable. Negative binomial regression can be used for over dispersed count data, that is when the conditional variance exceeds the conditional mean.

Results:

- Bangladesh has been reporting new cases of the coronavirus (COVID 19) every day since March 14, 2020. Bangladesh reported over 1.58 million confirmed cases as of October 31, 2020. Out of these 28052 were fatal
- The highest and lowest death cases are observed in the Dhaka district (7925) and in the Bandarban district (18) respectively among 64 districts.
- If total population in a district increases by 1% unit, then the risk of death occurring increases by 1.5 times.
- If the people of a district maintain sanitization that they use soap or detergent for handwashing then a 1% unit increases will occur the risk of death decreases by 0.01 times.
- When literacy rate in a district increases by 1% unit then the risk of death occurring decreases by 0.03 times.
- We found that the aged population above 65, population density, urban residence, and literacy rates are significantly correlated with the COVID-19 mortality

Fig.1: CONCEPTUAL FRAMEWORK

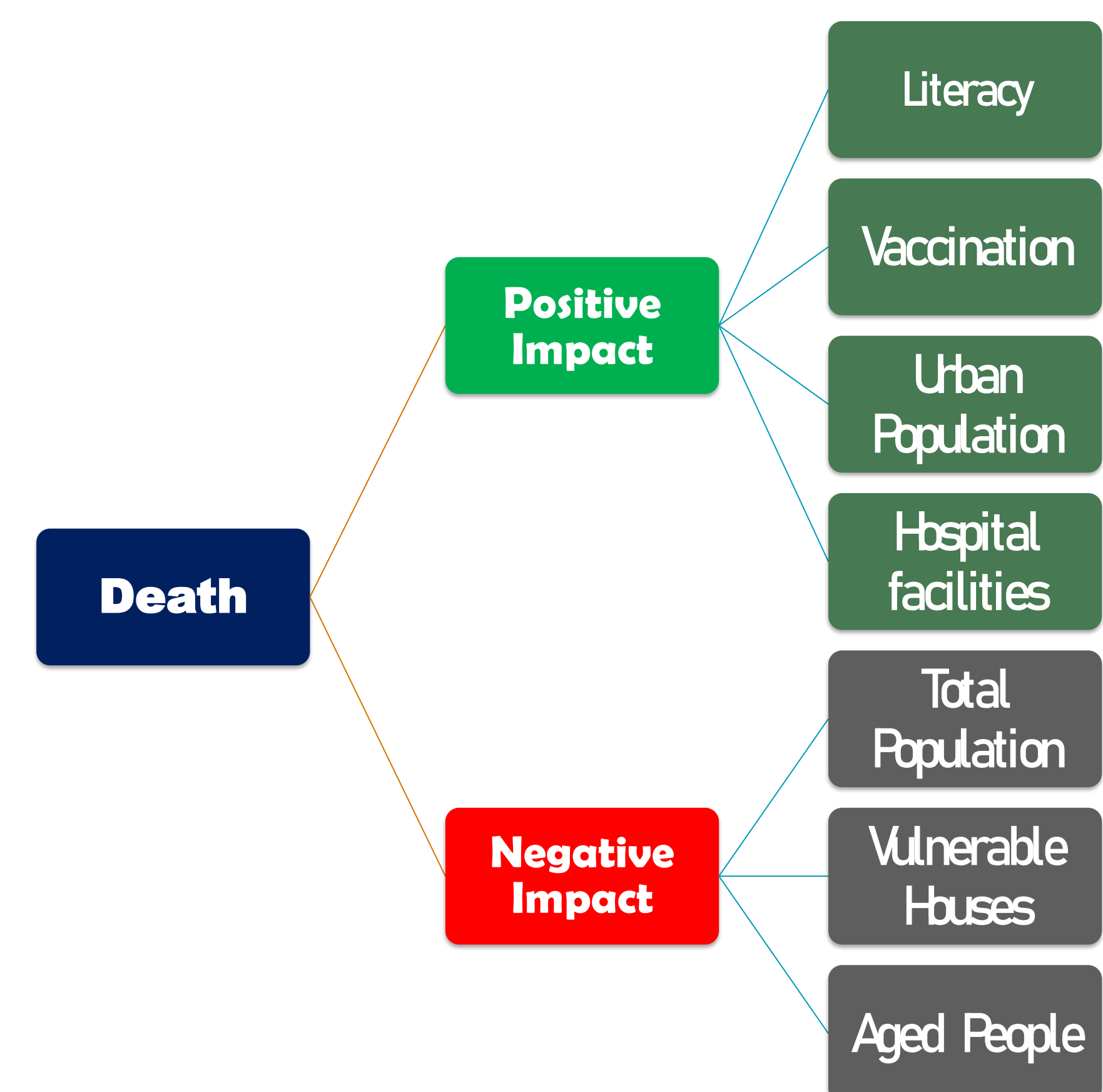


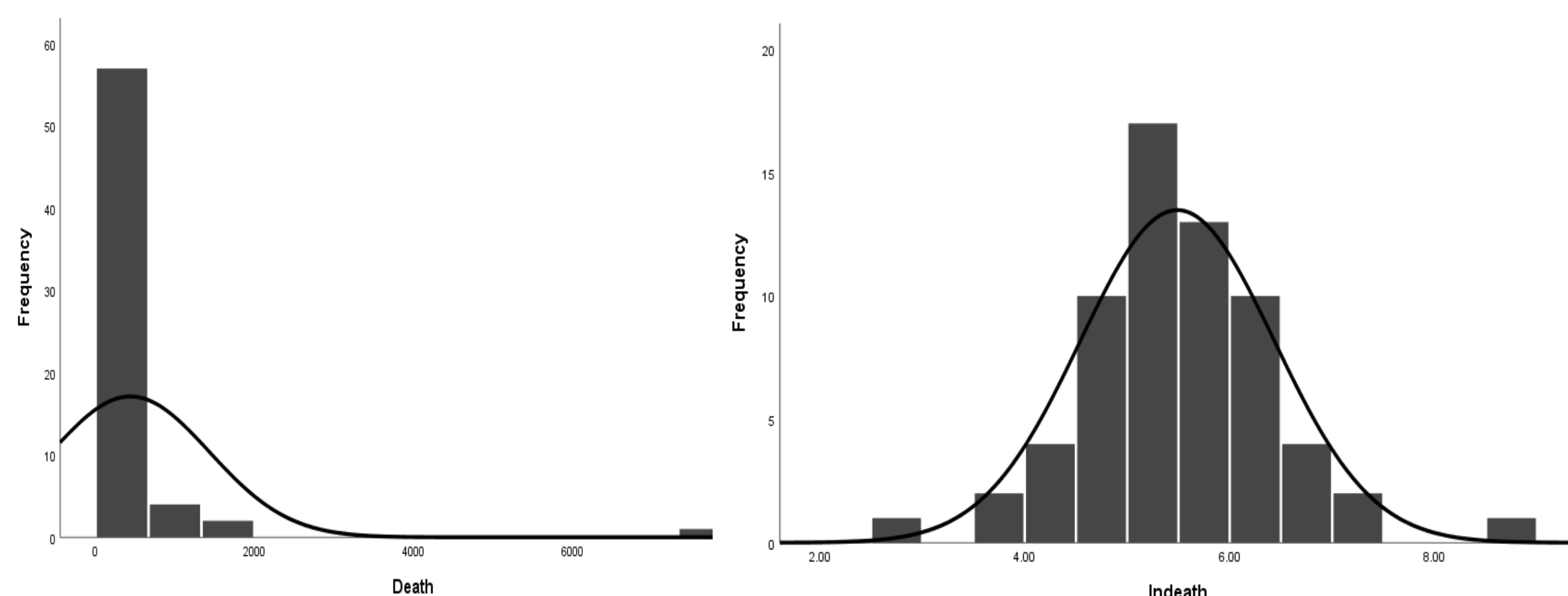
Table 1: DESCRIPTIVE ANALYSIS OF THE RISK FACTOS

Variable name	Minimum	Maximum	Mean	Std. Deviation	Skewness	Kurtosis
Vaccinated	21	44	29.05	5.314	0.692	0.147
Literacy Rate	34.98	70.54	50.17	7.65182	0.337	-0.116
Vulnerable House	0.13	2.4	0.665	0.4371	1.45	2.954
Hospital Facility for COVID patient	0.01	2.92	0.150	0.36905	6.999	52.373
Population	0.47	14.45	2.700	2.09111	3.428	16.284
Aged Population>65	2.68	6.62	5.001	0.82927	-0.705	0.598
Urban Population	0.25	27.76	1.562	3.55487	6.721	48.765
Soap or detergent available for handwashing at household	39.11	88.95	67.63	10.45891	-0.141	-0.027
Death	18	7925	438.3	993.966	7.027	53.1

Table 2: ANALYZE OF RISK FACTORS OF COVID-19 MORTALITY

Variables	Sig.	Exp (B)/IRR	CI
Vaccinated	0.565	0.990	0.955-1.026
Total population	0.000	1.500	1.306-1.722
Aged Population>65	0.877	1.017	0.822-1.259
Urban population	0.455	0.943	0.807-1.101
Soap or detergent available for handwashing at household	0.013	0.990	0.976-1.1003
Total beds available at hospitals for covid patients	0.802	0.861	0.268-2.764
Literacy rate	0.001	1.040	1.017-1.064
Vulnerable house	0.535	0.913	0.683-1.219

Graph:



Conclusions and Recommendation :

The COVID-19 pandemic is expected to have a long-term impact on health, economy, and social processes globally, including Bangladesh. The possible risk factors should be included in policy preparedness and implementation during the COVID-19 pandemic. Understanding risk factors of COVID-19 may also help to understand the future dynamics of COVID-19 or other such infectious diseases. The study findings provide crucial information for policy discourse, emphasizing the vulnerability of the highly urbanized and densely populated areas.